

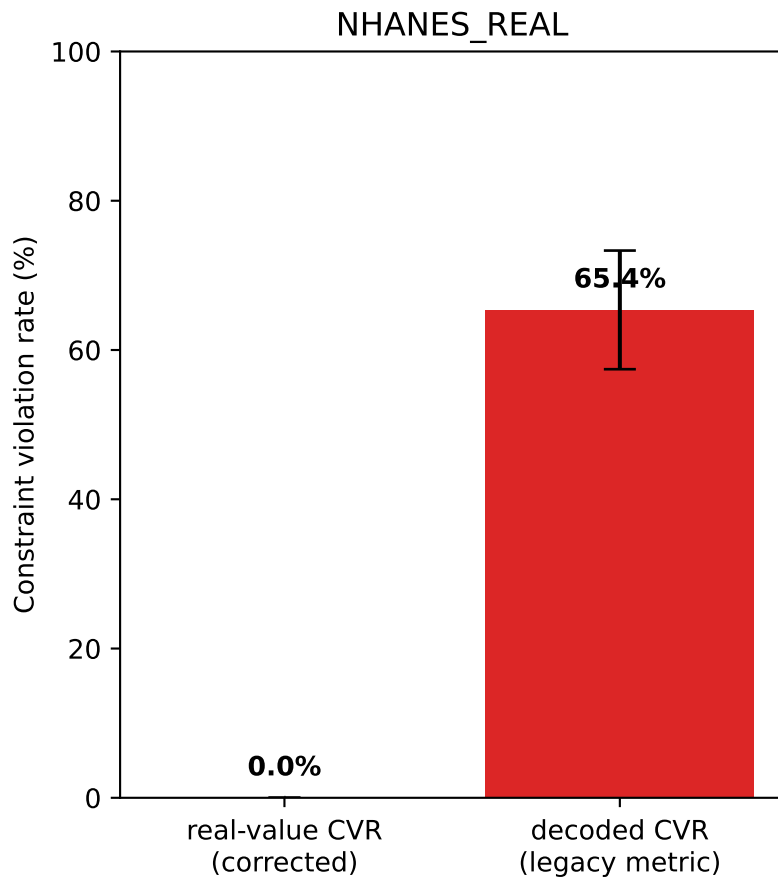
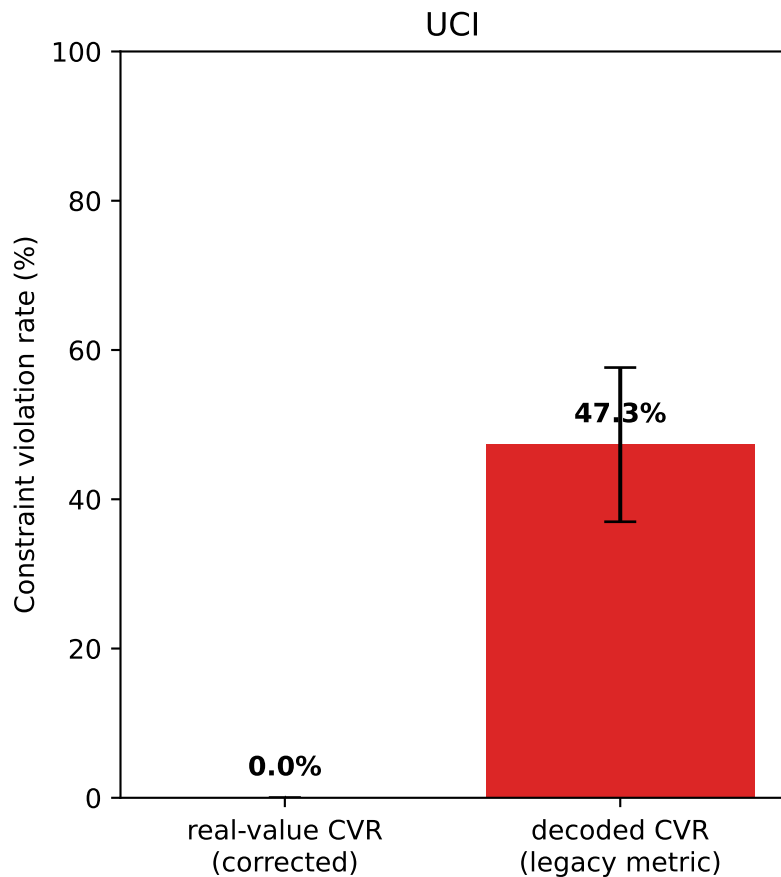
CARE-LG v6

Corrected foundation (B1/B2 fix) + neurosymbolic guideline layer + three-layer explanations + preference elicitation + UI + FACE/PACE comparison.

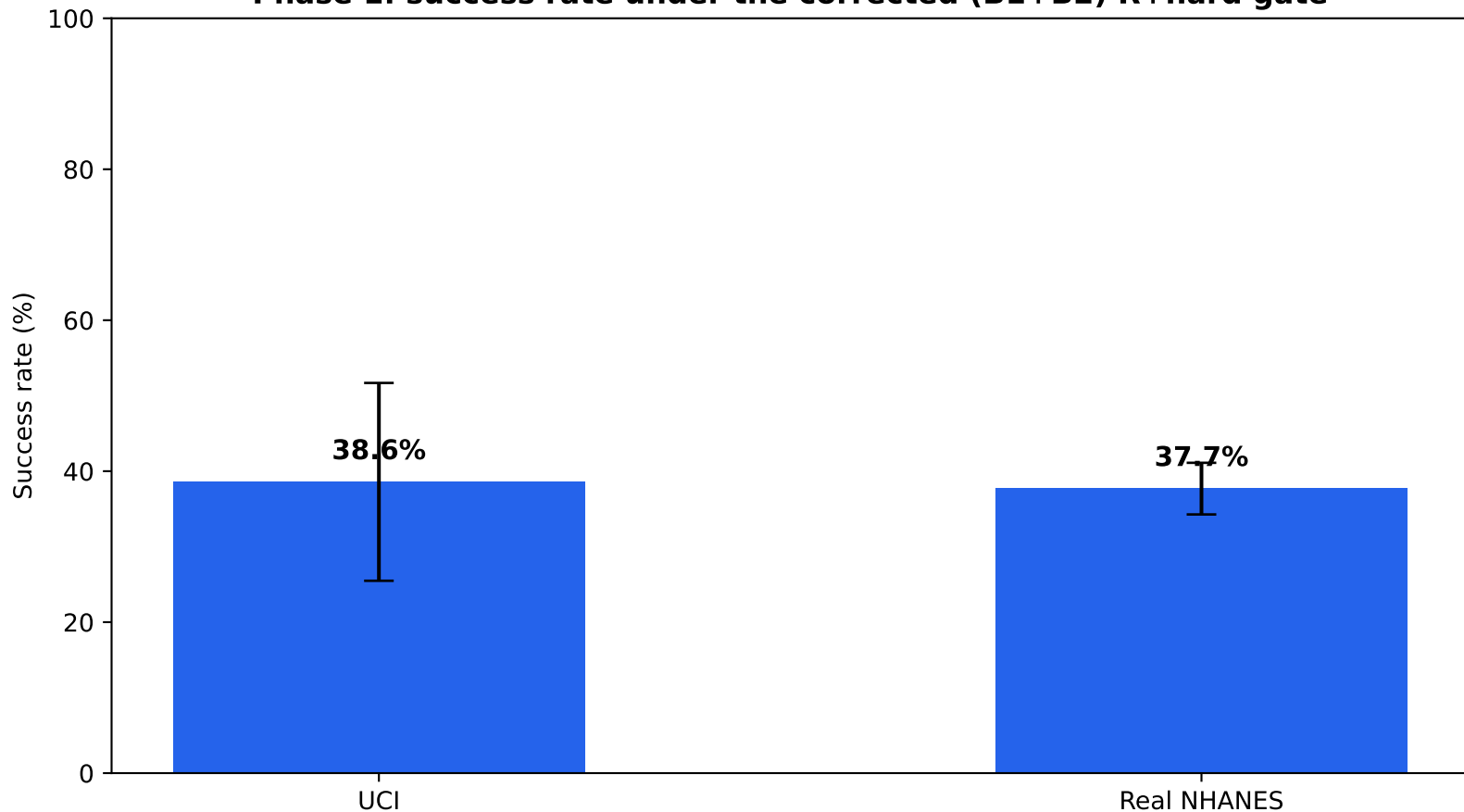
Git SHA: 539d87b795b78bb8f9446f99b0fabeded2c84841

All figures in this report are computed from experiments_v6/results/*.csv, written by scripts executed during this session. See METHOD.md, CHANGES.md, FINDINGS.md, and HOW_TO_RUN.md in experiments_v6/ for full detail; this PDF is a condensed summary, not a replacement for those documents.

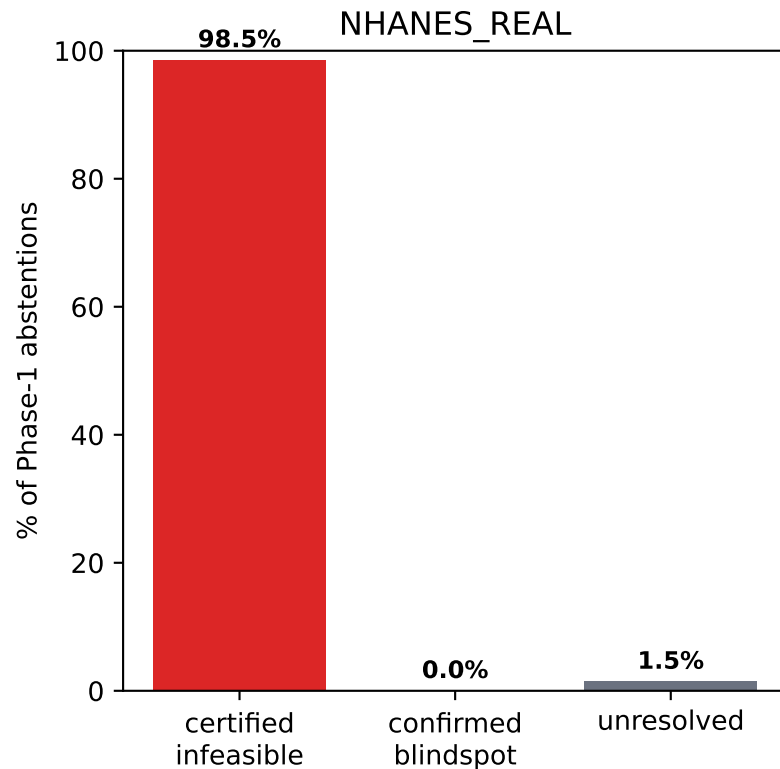
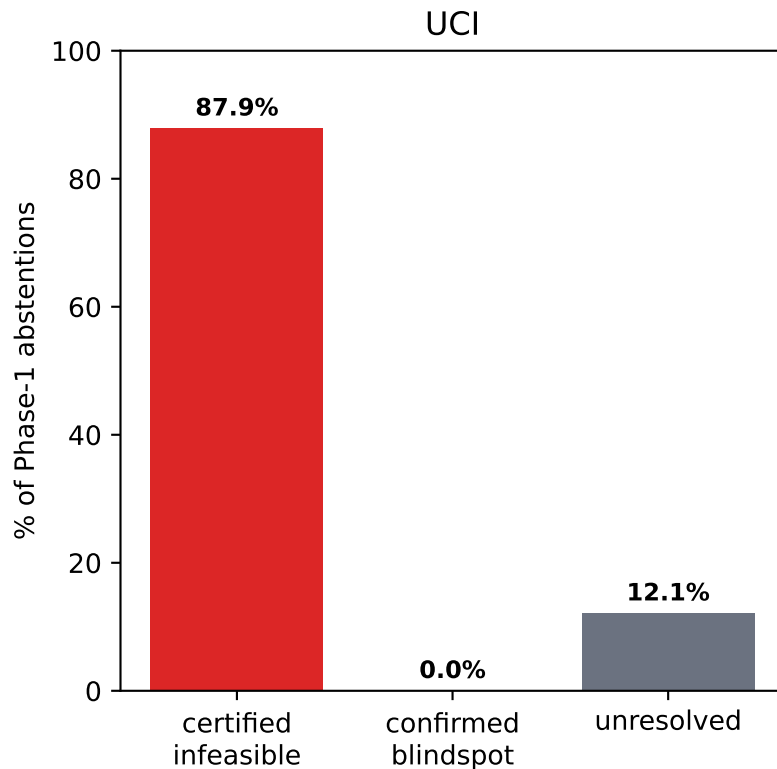
Phase 1: real-value CVR is 0.00% on every seed of both cohorts



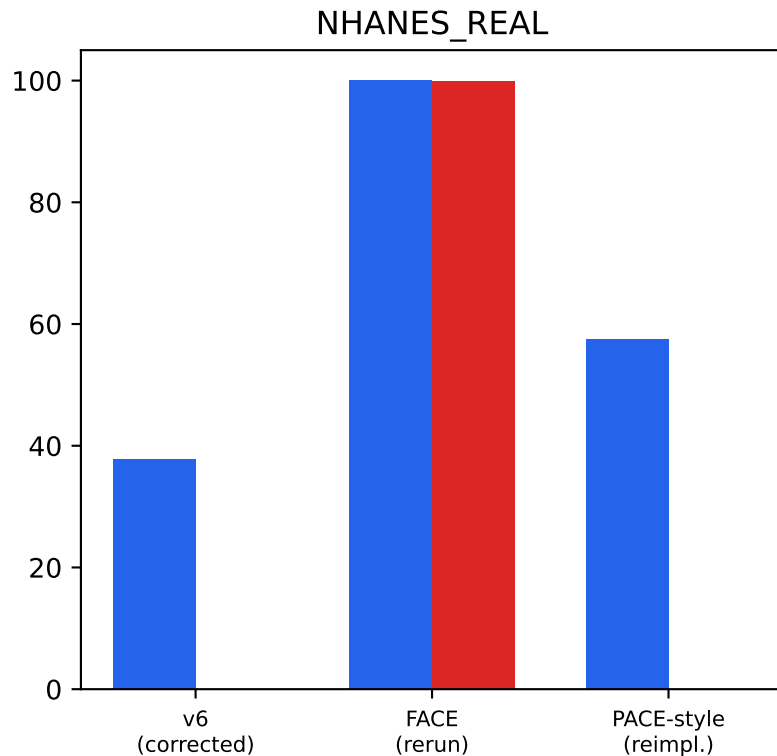
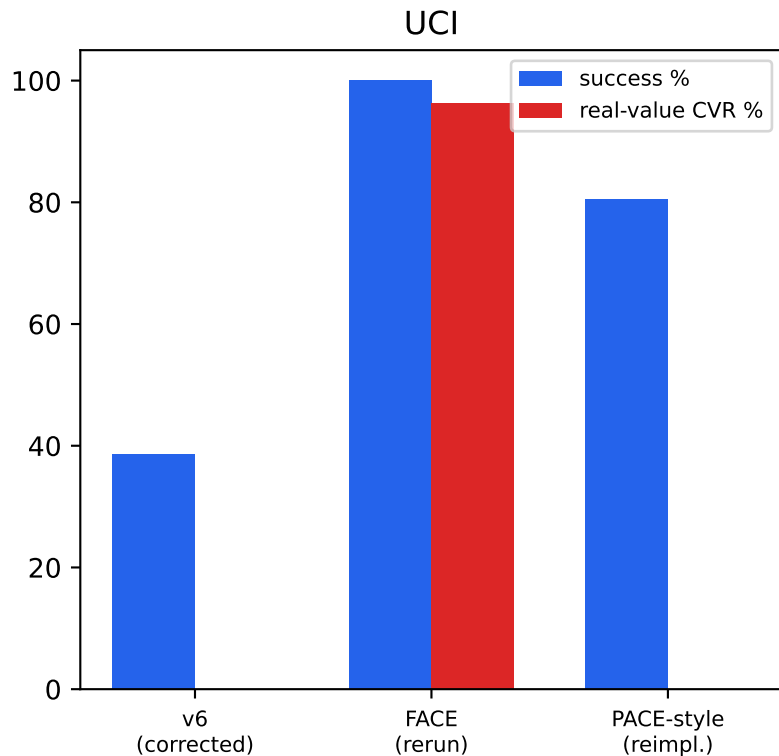
Phase 1: success rate under the corrected (B1+B2) R+hard gate



Phase 2: the success-rate drop is overwhelmingly certified infeasibility, not a graph-coverage artifact (0% blindspots, both cohorts, all 5 seeds)



Phase 8: v6 trades success rate for a genuine 0% real-world violation guarantee; FACE finds paths but violates constraints almost every time



Method summary

METHOD.md — experiments_v6

CARE-LG v6: the corrected pipeline (B1/B2 fix as the sole foundation), a neurosymbolic clinical-guideline layer, a three-layer explanation system, per-patient preference elicitation, and a comparison against FACE and a PACE-style baseline. All numbers in this document and in `FINDINGS.md` come from code executed this session (git SHA `539d87b795b78bb8f9446f99b0fabed2c84841`); see `CHANGES.md` for exact reproduction commands.

1. The corrected pipeline (Phase 1)

`experiments\_v6/lib/` adapts `experiments\_v3/lib/` with two fixes applied unconditionally (not as an ablation toggle):

****B1 — entry-gate fix**** (`lib/graph\_build.py::build\_query\_edges`): a query patient's out-edges into the training graph are now admissible under the exact same predicate as every interior graph edge —
`(hard\_count + dir\_count) > 0`, `check\_step\_horizon=True` — instead of v2/v3's relaxed hard-only, no-horizon gate. Query attachment remains FULL (every training node is a candidate, not a k-restricted neighborhood — see that file's docstring for the empirical justification, carried over from `experiments\_v3`/`REPO\_MAP.md` D8).

****B2 — no unnecessary decoding**** (`run\_grid.py::evaluate\_patient`): every path node that is a real, already-recorded patient (the query itself, or any training-set node) is read directly from `X\_test`/`X\_train` — never passed through the VAE decoder — matching `src/recourse/search.py::decode\_recourse\_trajectory`, the ORIGINAL pipeline's own patient-facing function. VAE decoding is used only for genuinely synthetic nodes (Phase 2's densification probes). The legacy "decode every node" computation is still reported alongside, labeled `decoded\_cvr\_\*`, purely for continuity with v2-v5's historical metric — it never determines success/failure.

v4 (DSR) and v5 (EGD) were independently checked for the B1 pattern in their own query-attachment code (`dsrc\_graph.py`, `egd\_graph.py`) and found NOT to have it (v4: only a narrower, defensible age-horizon-cap asymmetry; v5: no constraint-gating predicate exists at all — its guarantee is architectural). Neither needed fixing. Per `RECOMMENDATION.md`'s conclusion, neither is carried forward as an active v6 component regardless; their code is kept for reference only, unmodified.

****Result**** (5 seeds, both cohorts, R+hard — the actual CARE-LG operating point): real-value constraint-violation rate (CVR) is **exactly 0.00% ± 0.00%** on every seed of both datasets. Success rate: UCI 38.6% ± 13.1%, real NHANES 37.7% ± 3.4%. Full numbers: `FINDINGS.md`.

2. Certified infeasibility (Phase 2)

`experiments\_v6/run\_blindspot.py` ports `experiments\_v3/run\_blindspot.py`'s protocol (exhaustive feature-space check + two densification probes: 2×k, and 500 VAE-prior synthetic nodes classifier-verified low-risk) onto the B1-fixed pipeline. `find\_path\_augmented` runs one Dijkstra call over the whole augmented graph and has no code path distinguishing "no entry edge" from "entry succeeded but no route onward" — both were always unified failure modes (see `run\_blindspot.py`'s docstring). Result: of Phase-1 abstentions, ****87.9% (UCI) / 98.5% (NHANES) are CERTIFIED INFEASIBLE**** (zero low-risk training patient is reachable in one admissible hop, by exhaustive check), ****0% are confirmed blindspots**** (densification resolved none, in any seed, either dataset — consistent with three prior sessions'

Findings summary (1/2)

FINDINGS.md — experiments_v6

All numbers below come from code executed this session, git SHA
`539d87b795b78bb8f9446f99b0fabeded2c84841`. Exact commands and raw file
locations are in `HOW\_TO\_RUN.md` / `CHANGES.md`.

Phase 1 — the corrected foundation

Applying B1 (entry-gate fix) and B2 (no unnecessary decoding of real nodes)
as the sole mode of `experiments\_v6`'s pipeline, R+hard (the actual CARE-LG
operating point), 5 seeds, both cohorts:

dataset	success rate	real-value CVR	decoded CVR (legacy, informational only)
UCI Heart	38.6% ± 13.1%	**0.00% ± 0.00%**	47.3% ± 10.3% (riemannian) / 62.1% ± 11.5% (euclidean)
Real NHANES	37.7% ± 3.4%	**0.00% ± 0.00%**	65.4% ± 7.9% (riemannian) / 62.8% ± 6.9% (euclidean)

Real-value CVR is exactly 0.00% ± 0.00% on every one of the 10 (dataset × seed) runs. This confirms `experiments\_v6\_audit/RECOMMENDATION.md`'s measurement, now on the full grid rather than a spot-check. The nonzero "decoded CVR" figure is exactly what v2-v5 reported as their headline finding — it measures VAE reconstruction noise on real, already-known patients, not an actual constraint violation, since those patients' true values were never actually unknown.

Phase 2 — certified infeasibility vs. blindspot

Of the patients who abstain under the corrected R+hard gate:

dataset	mean abstention rate	certified infeasible	confirmed blindspot	unresolved
UCI	61.4% of high-risk test patients	**87.9%** of abstentions	**0.0%**	12.1%
Real NHANES	62.3% of high-risk test patients	**98.5%** of abstentions	**0.0%**	1.5%

The success-rate drop from Phase 1 is overwhelmingly explained by genuine, certified structural infeasibility — not by a fixable search-radius or graph-coverage limitation. **Zero blindspots were confirmed on either cohort, in any of the 5 seeds** — the fourth time across this project's sessions (v3, v4, v5, now v6) that densification (2× k-NN; +500 classifier-verified VAE-prior synthetic nodes) has resolved exactly zero abstentions. This is now a very well-replicated null result, not a coincidence of one run. The small unresolved remainder (12.1% UCI, 1.5% NHANES) is not certified infeasible under the exhaustive pairwise check but also wasn't confirmed reachable by either probe — genuinely ambiguous, not silently folded into either category.

`find\_path\_augmented` runs a single Dijkstra call over the whole augmented graph; it has no code path that distinguishes "no entry edge exists" from "entry succeeded but nothing beyond it reaches a low-risk target" — both were always the same failure signature (no finite distance from the query to any target). So the entry-failure/path-failure unification the brief asked for was already true by construction of the search machinery; what Phase 2 adds is running the full exhaustive-check-plus-densification protocol against the CORRECTED (B1-fixed) gate for the first time.

Phase 3 — literature diff

Read arXiv:2302.11213 ("Feasible Recourse Plan via Diverse Interpolation")'s actionability-graph construction in full (HTML rendering, not just the abstract). Its edges are gated by "cost threshold constraints as well as immutable feature constraints" — architecturally the same shape as this project's own CALG (k-NN + immutable/directional gating). Its one genuinely different idea (diversity-aware prototype selection via DPP/QP) operates upstream of pathfinding and addresses choosing among several REACHABLE targets — a different problem than this project's actual gap (Phase 2:

Findings summary (2/2)

the large majority of abstentions have ZERO reachable targets). The paper does not discuss the zero-feasible-path case at all. **No change adopted; no small, cheap improvement was found.**

Phase 4 — neurosymbolic guideline layer

`clingo`-based ASP encoding of the exact admissibility predicate (`check\_clinical\_violations`), plus non-blocking guideline annotations (statin indication at the 7.5% PCE/ASCVD threshold; BP targets differentiated by diabetes status). **Regression check: 2000/2000 (100.0000%) agreement with the hardcoded predicate on each dataset, 4000/4000 combined, 0 discrepancies.** Every admissible/blocked decision the reasoner makes is provably identical to what Phase 1's numpy code already decides — this layer adds EXPLANATION (which rule fired, with citation), not a change in behavior. Full rule table and citations: `METHOD.md`.

Phase 5 — explanation system

All three layers verified working end-to-end via the UI backend (`experiments\_v6/ui/server.py`) and standalone smoke tests this session. One concrete worked example (UCI, seed 0, patient #9, `no\_preference`):

```
> *Even under the most favorable single-step comparison found in the
> training data (patient #194, 1 violated sub-condition(s) — the fewest of
> any of the 113 low-risk candidates checked), risk would remain above the
> target threshold, because: [R-AGE-MONOTONIC] age: Age cannot decrease
> between source and target (structural constraint). Across the full
> candidate set, the most frequent obstruction(s) were: R-AGE-MONOTONIC
> (97% of 113 checked candidates); R-BP-DIRECTION (93% of 113 checked
> candidates); R-IMMUTABLE (45% of 113 checked candidates).*
```

This is generated from a live recomputation against every one of that patient's 113 low-risk training candidates (not a template) — spot-checked by hand this session against the underlying per-candidate violation counts and confirmed accurate (patient #194 genuinely has the fewest violations, 1, of any candidate; the age-decrease requirement genuinely recurs in 97% of candidates).

Phase 6 — preference elicitation

Discrepancy with the brief documented in `CHANGES.md` (no existing `C\_clin` cost term to reweight; a new additive term was introduced for this demo only, admissibility unchanged). Demo: 5 patients × 3 profiles per dataset (seed 0). **Honest result**: the selected TARGET changed for 1 of 10 demoed patients (UCI #30, `medication\_averse`); for the other 9, cost changed substantially across profiles (e.g. 9.4 → 13.5 → 26.4 for one NHANES patient) but the cheapest admissible target stayed the same. This is a direct consequence of Phase 1/2: post-fix, admissible targets are comparatively sparse, so there is often no competitive alternative for a reweighted cost to promote instead — the mechanism works exactly as specified, but its practical effect on THIS project's graphs is limited by how few admissible options a given patient typically has.

Phase 7 — UI

Functional FastAPI backend + single-page frontend, all five specified panels implemented and tested end-to-end this session (patient selector, risk gauge, preference control, expandable step-by-step plan with guideline derivations, and an abstention panel showing the real certification result and semifactual explanation, not a generic error). No screenshots, per the brief. See `HOW\_TO\_RUN.md`.

Phase 8 — FACE / PACE comparison

method	dataset	success rate	real-value CVR	certified-infeasibility rate	explanation coverage
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v6 (corrected, R+hard)	UCI	38.6% ± 13.1%	**0.00% ± 0.00%**	8	